University Important questions

Q1. Enumerate plasma proteins giving normal values.

Explain functions of plasma protein

Q2. Classify WBCs and describe their function with diagrams

Q3. What is the basis for classification of blood groups and what are the uses of blood grouping

Q4. Define clotting the time. Describe the mechanism of a coagulation of a blood Or Intrinsic and extrinsic pathway of blood coagulation

Q5. Immunity

Q6. Mention different antiquaguagulants and mechanism of action

Q7. What is erythropoises and where does it occur. Give stages of erythropoises and requirement of each stage

Q8. Define blood pressure. What is the normal blood pressure. Describe the short term regulation of blood pressure

Q9. Describe the long term regulation of blood pressure

Q10. What is hemoglobin. Describe the types, functions and fate of haemoglobin

Q11. Erythroblastosis foetalis

The most important questions are:

Q1, Q2, Q4, Q8, Q9, Q10

Q1. Enumerate plasma proteins giving normal values. Explain functions of plasma protein.

Plasma Proteins

Important Plasma Proteins

- **Albumins** - 48 g/L, 4.8 g%, **70,000** mol. wt.
- **Globulins** - 23 g/L, 2.3 g%, **21,000-150,000** mol. wt.
- **Fibrinogen** - 3 g/L, 0.3 g%, **340,000** mol. wt.
- **Normal A/G ratio** - **1.5:1**

Other Physiological Important Plasma Proteins

- **Prothrombin**
- **Angiotensinogen**
- **Complement system**
- **Kinin system**
- **Other blood clotting factors**
- **Fibrinolytic system**
- **Lipoproteins**
- **Haptoglobin**
- **Fetuin**
- **α-antitrypsin**
- **Proteins C and S**
- **Transport proteins**
  - **Transferrin**
  - **Ceruloplasmin**
  - **TBG, TBPA, CBG**

Albumin

- **Content** - 4-4.5 g%
- **Major plasma protein**
- **Molecular weight** - **69,000**

Globulin

- **Molecular weight** - **130,000**
- **Types** - **α, β, γ globulin**
- **β-globulin (Prothrombin)** - helps in coagulation
- **γ-globulin** - forms **antibodies**

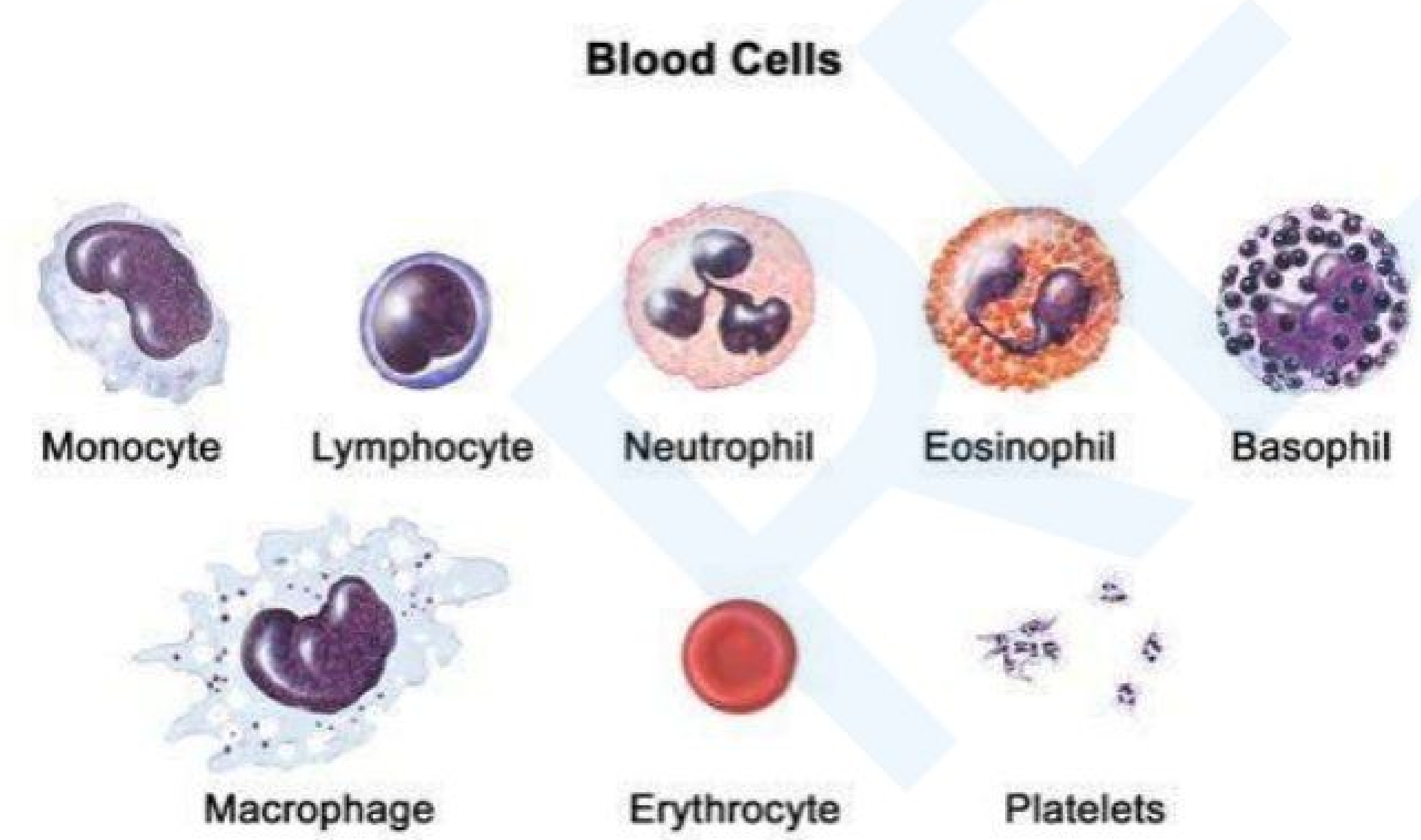
Fibrinogen

- Required for **coagulation**
- Converts to **fibrin** during clot formation
- **Prothrombin & fibrinogen** used in clotting

Functions of Plasma Proteins

- **Colloidal Osmotic Pressure** - **25 mmHg**, prevents fluid loss, **Albumin** major contributor
- **Protective Function** - **γ-globulins** form **antibodies (IgA, IgG, IgD, IgE, IgM)**
- **Coagulation** - **Fibrinogen, Prothrombin, Antihaemophilic globulin** involved in clotting
- **Transport** - **Albumin, α & β-globulins** transport **CO2, hormones, metals, drugs**
- **ESR (Erythrocyte Sedimentation Rate)** - influenced by **fibrinogen**
- **Viscosity** - **Plasma proteins + RBCs = 50% each**, **Fibrinogen & Globulins** contribute
- **Buffer Mechanism** - **Acid-base balance** maintained by plasma proteins
- **Protein Reserve** - Used in **starvation**, broken down into **amino acids** for body protein synthesis

Q2. Classify WBCs and describe their function with diagrams.



White Blood Cells (Leucocytes)

General Features

- **Largest formed elements** in blood **larger than RBCs** contain **nucleus**

- **Normal count** - **4000-11,000** cells/cu mm blood **Children** - **18,000-25,000** cells/cu mm
- **Life span** - **Few hours to 2-3 days**

Types of Leucocytes

- **Granulocytes** (contain granules)
- **Agranulocytes** (do not contain granules)

Granulocytes

Neutrophils

- **60-70% of total WBCs**
- **Multilobed nucleus** (3-5 lobes) **Fine granules neutral stain**
- **Phagocytosis** increases in **acute infections** (pneumonia, appendicitis, tonsillitis, abscess)
- **Increases** in **menstruation, pregnancy, exercise**
- **Granules contain enzymes** - **myeloperoxidase, proteases** destroy microorganisms
- **Release thromboxanes, leukotrienes** increase **vascular permeability** and migration of neutrophils
- **Reduced count (Neutropenia)** - **Typhoid, malaria, aplastic anemia, drug effects**

Eosinophils

- **Bilobed nucleus Large granules acidophilic stain**
- **Normal count** - **2-5%**
- **Increase (Eosinophilia)** - **Allergic conditions, parasitic infestations**
- **Decrease (Eosinopenia)** - **Acute pyogenic infections, ACTH/steroid therapy**
- **Mild phagocytosis**

Basophils

- **Bilobed nucleus Coarse blue granules**
- **Normal count** - **0-1%**
- **Releases heparin (anticoagulant) & histamine** (hypersensitivity reactions like anaphylactic shock)
- **Increase** in **polycythemia, chronic myeloid leukemia**

Agranulocytes

Lymphocytes

- **Large round nucleus Thin strip of cytoplasm**
- **Count** - **20-30%** increases in **chronic infections (tuberculosis)**
- **Provide immunity**
- **Two types** - **T & B lymphocytes** involved in **immune defense**

Monocytes

- **Largest WBCs (15 μm) Kidney-shaped nucleus**
- **Normal count** - **3-8%**
- **Phagocytic function** - **Second line of defense**
- **Enter tissues** as **macrophages**

Functions of Leucocytes

- **Phagocytosis** - **Neutrophils, monocytes engulf bacteria & foreign material**
- **Antiallergic effect** - **Eosinophils inhibit histamine release**
- **Antibody formation** - **Lymphocytes produce antibodies for immunity**
- **Heparin production** - **Basophils prevent intravascular clotting**
- **Trephine formation** - **Leucocytes help in tissue growth & repair**

Q3. What is the basis for classification of blood groups and what are the uses of blood grouping.

BLOOD GROUP SYSTEMS

1. ABO SYSTEM

- Based on the presence of **antigens** A and B on **RBC** membrane.
- Four **blood groups**: A, B, AB, O.
- **A antigen** has two subtypes: **A1** and **A2**, resulting in six possible blood groups: A1, A2, B, A1B, A2B, and O.
- **Karl Landsteiner's Law**:
  - If **agglutinogen** (**antigen**) is present on RBCs, **agglutinin** (**antibody**) is absent in plasma.
  - If agglutinogen is absent, the corresponding agglutinin will be present.
- **Blood Group Combinations**:
  - Group A: **A antigen**, **anti-B antibody**.

- Group B: **B antigen**, **anti-A antibody**.
- Group AB: **A** and **B antigens**, **no antibodies**.
- Group O: **No antigens**, **anti-A** and **anti-B antibodies**.

2. RH SYSTEM

- Discovered by **Landsteiner** and **Weiner** in 1940.
- **Rh+ve** individuals show **agglutination** with rabbit serum (**85%**), while **Rh-ve** show **no agglutination** (**15%**).
- There are three **antigens**: **C**, **D**, and **E,** with **D** being the most important.
- **Genotypes**:
  - **DD**, **Dd** are **Rh+ve,** and **dd** is **Rh-ve.**
- **Rh Antibodies** form only when **Rh-ve** individuals are exposed to **Rh+ve blood** or fetal RBCs.
- **Rh Distribution**:
  - **Caucasians**: **85% Rh+ve**, **15% Rh-ve**.
  - **Orientals**: **95% Rh+ve**, **5% Rh-ve**.

3. SIGNIFICANCE OF RH GROUPING

- **Transfusion**: Rh+ve blood should not be given to **Rh-ve** individuals to prevent antibody formation.
- **Pregnancy**:
  - If the **father is Rh+ve**, **mother is Rh-ve**, and **fetus is Rh+ve**, the mother's body can produce antibodies during delivery.
  - In subsequent pregnancies, these antibodies may cross the placenta, causing **hemolysis** of the fetus (**HDN**).
  - **HDN Symptoms**: **Jaundice**, **erythroblastosis fetalis**, **hydrops fetalis**, and **hemolytic anemia**.
  - The frequency of HDN in second or later pregnancies: **1 in 22**.

4. PREVENTION OF RH COMPLICATIONS

- If a **Rh-ve mother** has an **Rh+ve fetus**, **Anti-D antibodies** are given immediately after delivery to prevent complications in future pregnancies.
- **Fetal Protection**: In some cases, **Rh-ve blood** can be infused into the fetus to avoid the anti-Rh antibodies from attacking the fetus.

5. MNS SYSTEM

- Mainly used in **medicolegal** cases, not routine clinical practices.

6. IMPORTANCE OF BLOOD GROUPS

- **Transfusion**: To avoid reactions during **blood transfusions**.
- **Medicolegal**: Helpful in **disputed parentage** and **criminal cases** to exclude potential fathers.
- **Research**: Blood groups can be linked to higher risk for certain **diseases**.
- **Anthropology & Ethnology**: Blood groups help in understanding **population genetics**.
- **Organ Transplantation**: Matching **blood groups** is crucial to avoid organ rejection.
- **HDN Prevention**: **Rh typing** is important to avoid **hemolytic disease** in newborns.

Q4. Define clotting the time. Describe the mechanism of a coagulation of a blood Or Intrinsic and extrinsic pathway of blood coagulation.

Clotting Time

- The time taken by blood to form a **clot** after being exposed to air or injury.
- It reflects the efficiency of the **coagulation process** and the functioning of **platelets** and **clotting factors**.

COAGULATION

- **Coagulation** is the process where blood changes from a **fluid** to a **gelatinous or semisolid** state due to injury to blood vessels.
- This change is accompanied by **physicochemical changes** in the blood.

Stages of Coagulation

Stage 1: Formation of Prothrombin Activator

- Inactive **procoagulants** become **active**, leading to the formation of **prothrombin activator**.

Stage 2: Conversion of Prothrombin to Thrombin

- **Prothrombin** is converted to **thrombin**.

Stage 3: Conversion of Fibrinogen to Fibrin

- **Fibrinogen** is converted into **fibrin** to form the clot.

Process of Clotting

- **Clotting factors** are **proenzymes** activated during clotting. Each step amplifies the reaction, known as the **cascade mechanism**.
- Injury to blood vessels exposes **collagen** and the **negative charge** on the endothelial surface, leading to:
    - **Platelet adhesion** and aggregation, forming a temporary **haemostatic plug**.
    - Release of **platelet contents** (e.g., ADP, vasoconstrictors, **platelet factor** III), which activate the clotting process.

Generation of Prothrombin Activator

- The first step is the formation of **activated factor X**, also known as **prothrombin activator**.
    - **Intrinsic pathway** (within the blood).
    - **Extrinsic pathway** (from tissues).

Intrinsic Pathway

- **Platelets** in the blood initiate the formation of **prothrombin activator**.
- **Steps**:
    - Injury to blood vessel → release of **thromboplastin** (**Factor** III).
    - **Activation of Factor** VII **to VIIa** → activates **Factor X to Xa**.
    - **Factor** I**Xa** forms **prothrombin activator**.

Extrinsic Pathway

- Involves **tissue thromboplastin**, **calcium ions**, and **Factor V**.
- **Steps**:
    - Injury to blood vessel → release of **phospholipid** from damaged cells.
    - **Activation of Factor** VII → activates **Factor X** to **Xa**.
    - **Activated Factor Xa** leads to **formation of prothrombin activator**.

Formation of Fibrin

- Blood coagulation involves converting **fibrinogen** (soluble plasma protein) into **fibrin** (insoluble clot) by **thrombin**.
- Initially, **fibrin** forms as a **soluble monomer**.
- **Factor** XIII **(Fibrin-stabilizing factor)**, in the presence of **thrombin**, cross-links **fibrin threads** to form **fibrin polymer**.
- **Calcium ions** are required for **fibrin polymerization**.
- The final **fibrin polymer** traps **red cells** and **serum**, forming the **insoluble clot**.

Q5. Immunity

Immunity

- The body's defense system against harmful **microorganisms** and **foreign substances**
- Includes **Innate immunity** and **Adaptive immunity**

Innate Immunity

- First line of defense, **present from birth**
- **Non-specific** response involving **skin**, **mucous membranes**, **phagocytes**, and **complement system**
- **Natural killer (NK) cells** kill **infected** or **abnormal cells**
- **Inflammation** helps isolate and eliminate pathogens

Adaptive Immunity

- **Specific** and **acquired** response after exposure to pathogens
- Involves **B-cells** and **T-cells**

Humoral Immunity

- **B-cells** produce **antibodies** that target **antigens**
- **Plasma cells** secrete antibodies, while **memory B-cells** provide long-term immunity

Cell-Mediated Immunity

- **T-cells** directly target **infected** or **abnormal cells**
- **Helper T-cells** activate B-cells and cytotoxic T-cells
- **Memory T-cells** provide long-term protection

Antigen-Presenting Cells (APCs)

- **Dendritic cells**, **macrophages**, and **B-cells** present **antigens** to activate T-cells

Cytokines

- **Interleukins**, **interferons**, and **tumor necrosis factors** regulate immune responses

Immunological Memory

- **Memory cells** remember previous pathogens and respond faster on re-exposure

Vaccination

- Stimulates adaptive immunity without causing disease by introducing **antigens**

Immunodeficiency

- Weakening of the immune system, leading to increased susceptibility to infections

Autoimmunity

- Immune system mistakenly attacks the body's own tissues, like in **rheumatoid arthritis**

Hypersensitivity

- Overreaction of the immune system, such as in **allergies** or **delayed-type hypersensitivity**

Q6. Mention different antiquaguagulants and mechanism of action.

Salts

- In **extrinsic pathway**, the formation of **prothrombin activator** is initiated by **tissue thromboplastin**.

Chelating Agents

- **Chelating agents** such as **citrates**, **oxalates**, **fluorides**, and **EDTA (ethylenediaminetetraacetic acid)** bind to **calcium**.
- **Calcium** is required for many steps in the **coagulation process**, and its removal prevents **clotting**.

Heparin

- **Heparin** is a naturally occurring **anticoagulant** secreted by **mast cells** and **basophils**. It is also present in tissues like **lungs** and **liver**.
- It is a **negatively charged sulphated polysaccharide** and requires **antithrombin factor** III for its **anticoagulant** activity.
- Heparin combines with **heparin cofactor** and inhibits the action of **thrombin**. It also blocks the formation of **prothrombin activator** by inhibiting **factor IX**.
- **Heparin** is used widely in clinical practice and laboratories as an **anticoagulant** both **in vivo** and **in vitro**.

- Its action can be **inhibited** by **protamine sulphate**, a **positively charged polypeptide** obtained from **fish sperm**.

Dicoumarol

- **Dicoumarol** is a **Vitamin K antagonist**. It competitively inhibits the action of **vitamin K** in the liver.
- **Vitamin K** is required for the synthesis of **prothrombin**, **factors VII**, **IX**, and **X**.
- Administration of **dicoumarol** blocks the synthesis of these **clotting factors**.
- **Dicoumarol** is effective as an **anticoagulant** only **in vivo**.

Q7. what is erythropoises and where does it occur. Give stages of erythropoises and requirement of each stage.

Erythropoiesis

- **Erythropoiesis** is the process by which mature **RBCs** are produced from precursor **stem cells**.
- **Time required**: **5-10 days**.

Important Sites of Erythropoiesis:

1. **Mesoderm of yolk sac** → during first 3 months of **fetal life**.
2. **Liver and spleen** → during 3-6 months of **fetal life**.
3. **Red bone marrow** → in the rest of fetal life and in postnatal period:
    - From **birth to adulthood**, the entire **bone marrow** is involved in blood cell production.
    - From **adulthood onwards**, only **red bone marrow** is capable of producing blood cells.
    - In adults, **yellow fatty tissue** occupies the shaft of long bones, while **red bone marrow** at the ends of long bones, flat bones like **skull**, **sternum**, **vertebrae**, **pelvis**, and **ribs** are active in blood cell production.
    - **Liver and spleen** lose the function of producing blood cells in adults but can still produce cells during **extramedullary haemopoiesis** if stimulated.

Stages of Erythropoiesis

1. **Haemocytoblast (Stem Cell)**
2. **Proerythroblast**
3. **Early normoblast**

4. **Intermediate normoblast**
5. **Late normoblast**
6. **Reticulocyte**

Haemocytoblast (Stem Cell)

- **Uncommitted and pluripotent** cell that can give rise to either **erythroid** or **myeloid** series.
- 75% of bone marrow cells belong to **myeloid series**, while 25% belong to the **erythroid series**.
- These stem cells give rise to **committed stem cells** and **multipotent cells** to maintain constant numbers.

Proerythroblast

- A **large precursor cell** (18-20 μm in size) with a **single nucleus** and large nucleolus.
- Shows **active proliferation**.

Early Normoblast

- Cell size reduces to **16-18 μm**.
- The **nucleus condenses**, and **chromatin threads** appear.
- Nucleoli disappear, and the cell continues **active proliferation**.

Intermediate Normoblast

- **Nucleus further condenses** and cell size reduces to **12-14 μm**.
- The cytoplasm shows **haemoglobin synthesis**, taking both **acidic** and **basic** stains (**polychromatophilic**).
- **Cell division stops** at this stage.

Late Normoblast

- The cell size reduces to **10-12 μm**.
- **Nucleus pushed to periphery**, and it is extruded by **pyknosis**.
- Greater **haemoglobin synthesis** in cytoplasm.

Reticulocyte

- After the extrusion of nucleus, the **reticulocyte** is formed and released into the circulation.

- **Cell size** becomes **7.2 μm**.
- The cytoplasm has a **fine reticulum**, the nuclear remnant of **RNA.**
- Within **24-48 hours**, **haemoglobin synthesis** is completed and they mature into **red cells** in circulation.
- **Normal count** of reticulocytes is **1%** of red cell count.

Erythrocyte

- Both **reticulocytes** and **erythrocytes** are present in **peripheral circulation**.
- **Red cell** formed from **reticulocyte** exhibits its **normal morphology**.

Regulation of Erythropoiesis

- RBCs transport O_2, normal levels must be maintained
- **1%** of RBCs are destroyed daily, complete turnover every **4 months**

Factors

1. **Lack of O_2**: Stimulates erythropoiesis
2. **Nutrients**:

- **Carbohydrates**: Energy
- **Fats**: Cell membranes
- **Proteins**: Globin, enzymes, stroma proteins

3. **Minerals**:

- **Iron**: Heme in Hb, deficiency causes anaemia
- **Copper**: Iron transport, deficiency causes anaemia
- **Cobalt**: For Vitamin B_{12} action
- **Manganese**: For Hb synthesis

4. **Vitamins**:

- **Vitamin B_{12}** and **folic acid**: DNA synthesis
- **B_{12}** needs intrinsic factor for absorption

5. **Hormones**:

- **Erythropoietin (EP)** stimulates RBC production
- **Thyroid hormones**: Metabolism and RBC maturation
- **Adrenal cortex**, **sex hormones**: Influence EP secretion

6. **Neural**:

- **Hypothalamus stimulation** increases RBC production

Q8. Define blood pressure. What is the normal blood pressure. Describe the short term regulation of blood pressure.

Blood Pressure

Definition

- **Lateral pressure** exerted by **blood** on the **walls of blood vessels**
- **Driving force** for **circulation** of blood

Normal Blood Pressure

- **Systolic Pressure** - **100-140 mmHg** (Avg **120 mmHg**)
- **Diastolic Pressure** - **60-90 mmHg** (Avg **80 mmHg**)
- **Pulse Pressure** - Difference between **systolic** & **diastolic** (**40 mmHg**)
- **Mean BP** - **Diastolic Pressure + 1/3 Pulse Pressure**

Regulation of Blood Pressure

Short-Term Regulation

- **Immediate adjustment** through **reflex mechanisms**
- **Vasomotor Centre** - Located in **IV ventricle**, has **pressor** & **depressor areas**
- **Sympathetic Tone** - Maintains **partial constriction** of vessels through **thoracolumbar outflow**
- **Higher Centres** - **Cerebral cortex, hypothalamus, limbic lobe** influence BP

Sinoaortic Reflex

- **Baroreceptors** in **carotid sinus** & **aortic arch** detect **BP changes**
- **Carotid Sinus** - Supplied by **sinus nerve** (Glossopharyngeal)
- **Aortic Arch** - Supplied by **vagus nerve**
- **Increased BP** → Stimulates receptors → **Impulses** to **vasomotor & cardioinhibitory centres** → **↓ BP & ↓ Heart Rate**
- **Decreased BP** → Reverse changes
- **Buffer Nerves** - **Sinus Nerve & Vagus Nerve** help regulate BP

Other Reflexes

- **Left Ventricular Reflex** - **↑ LV pressure → ↓ BP**
- **Pulmonary Reflex** - **Lung congestion → ↓ HR & ↓ BP**
- **Coronary Chemoreflex** - **Veratridine injection** in **left coronary artery** → **↓ BP**

- **Cushing's Reflex** - **↑ Intracranial Pressure → ↑ BP & Bradycardia**
- **Higher Centre Reflex** - **Emotional stress → ↑ Sympathetic System → ↑ BP & HR**

Hormonal Regulation

- **Catecholamines, Serotonin, Angiotensin** II → **Vasoconstriction → ↑ BP**

Q9. Describe the long term regulation of blood pressure.

Long-Term Regulation

- Involves **kidneys**, **nervous system**, and **hormonal control** mechanisms
- The kidneys play a central role in regulating blood pressure through **fluid balance** and **electrolyte regulation**
- **Renin-Angiotensin-Aldosterone System (RAAS)**:
    - When blood pressure drops, kidneys release **renin**, which converts **angiotensinogen** into **angiotensin** I
    - Angiotensin I is converted to **angiotensin II** in the lungs by **angiotensin-converting enzyme (ACE)**
    - **Angiotensin** II constricts blood vessels and stimulates **aldosterone** release from the adrenal glands
    - Aldosterone increases **sodium** reabsorption in the kidneys, leading to **water retention**, increasing blood volume and pressure
- **Antidiuretic Hormone (ADH)**:
    - Released from the pituitary gland in response to low blood volume or high osmolarity
    - Promotes water reabsorption in the kidneys, increasing blood volume and pressure
- **Baroreceptor Reflex**:
    - Detects changes in blood pressure through baroreceptors in the **carotid sinus** and **aortic arch**
    - Signals sent to the brainstem to adjust heart rate and blood vessel tone, providing immediate regulation
- **Sympathetic Nervous System**:
    - Activates during stress or low blood pressure to release **norepinephrine**, increasing heart rate and vasoconstriction to raise blood pressure
- **Atrial Natriuretic Peptide (ANP)**:
    - Released from the heart in response to high blood volume
    - Promotes **sodium excretion** and **vasodilation**, helping to lower blood pressure
- Over time, these mechanisms adjust blood volume and vascular resistance to maintain stable **long-term blood pressure regulation**.

Q10. What is hemoglobin. Describe the types, functions and fate of haemoglobin.

Haemoglobin

Definition

- **Red pigment** in **RBCs**
- **Chromoprotein** with **mol. wt. 645,000**

Normal Concentration

- **Adult Male** - **14-18 g%**
- **Adult Female** - **12-16 g%**
- **Children** - **18-22 g%**

Functions

- **Carries O_2 & CO_2**
- **Acid-base regulation**
- **Transports NO (Nitric Oxide) → Vasodilation**

Structure of Haemoglobin

- **2 Parts** - **Heme** & **Globin**
- **4 Subunits** - Each with **1 Heme + 1 Globin**
- **Hb Molecule** - **4 Protoporphyrin IX + 4 Fe^{2+} + 4 Polypeptide chains**
- **Heme** - **Protoporphyrin + Iron**
- **Globin** - **4 Polypeptide Chains** (Each binds to **1 Heme**)
- **Adult Hb** - **2 α + 2 β Chains**
- **Fetal Hb** - **2 α + 2 δ Chains → More O_2 affinity**

Types of Haemoglobin

- **Hb A (Adult)** - **2 α + 2 β Chains**
- **Hb F (Fetal)** - **2 α + 2 δ Chains → Replaced by Hb A after birth**
- **Thalassemia** - **Defective α or β Chain synthesis**
- **Sickle Cell Hb (Hb S)** - **Mutant gene → Sickle-shaped RBCs → Osmotic resistance to Malaria**

Fate of Haemoglobin

- **RBCs destroyed** in **Reticuloendothelial system**
- **Hb released** as **Choleglobin**
- **Converted to Biliverdin** (**Green color**)
- **Biliverdin → Bilirubin** (**Yellow color**)

- **Bilirubin binds** to **protein** in **plasma**
- **In Liver** - **Conjugated with Glucuronic Acid**
- **Bilirubin excretion**
  - **Bacteria in intestines** → **Bilinogens & Stercobilinogen** (**Faeces**)
  - **Some reabsorbed** → **Urobilinogen** (**Urine**)

Q11. Erythroblastosis foetalis.

- A condition where the **fetal red blood cells** are destroyed by the **mother's antibodies**
- Usually occurs in **Rh-incompatible pregnancies**
- If the mother is **Rh-negative** and the fetus is **Rh-positive**, the mother's immune system may recognize the Rh antigen as foreign
- During pregnancy or delivery, **fetal Rh-positive blood** can mix with the mother's Rh-negative blood, triggering an **immune response**
- The mother's immune system produces **IgG antibodies** against the Rh antigen
- These antibodies can cross the **placenta** and attack the **fetal red blood cells**
- This causes **hemolysis** (destruction of red blood cells) in the fetus, leading to **anemia**
- The body responds by producing **immature red blood cells** (erythroblasts) to compensate for the loss
- **Symptoms** of erythroblastosis foetalis include **jaundice**, **hepatosplenomegaly**, and **hydrops fetalis** (severe edema)
- **Prevention**: **Rh immunoglobulin (RhIg)** is given to the mother after delivery or any event where fetal and maternal blood may mix, preventing the formation of antibodies
- **Treatment**: Severe cases may require **intrauterine blood transfusion** or early **delivery** of the fetus